

DiabetesAI

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Abstract

Diabetes is among the most prevalent chronic diseases in the United States, significantly affecting millions each year. This condition, in which one's body has trouble regulating its blood sugar levels, can lead to severe long-term complications if not properly managed. Early identification and intervention are crucial to prevent future complications. Despite the availability of lifestyle changes and medical strategies to manage diabetes today, many individuals remain unaware of their risk, which can lead to severe complications down the road and a reduced quality of life.

DiabetesAI aims to address these challenges by developing a comprehensive system that predicts diabetes risk and offers personalized health suggestions. Using advanced machine learning techniques, this model analyzes data, including health-related risk behaviors, chronic health conditions, and access and use of health-care to accurately assess the risk of diabetes. In addition, it provides suggestions to help people with diabetes improve their health. By training the model on non-diabetic people to establish baseline ranges for key indicators to detect diabetes such as BMI or GenHlth, it provides recommendations to diabetic individuals on how to improve their health and reduce risks. So, through its focus on improving early detection rates and health suggestions, DiabetesAI aims to improve overall health for people with diabetes.

Keywords

Diabetes, RandomForestClassifier, Machine Learning, SVM, Logistic Regression, Predictive Modeling, Diabetes Risk, Health Data Analysis

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1 Introduction

Diabetes is among the most prevalent chronic diseases in the United States, significantly affecting millions each year[3]. In 2021, the CDC reported that 38.4 million Americans were diagnosed with diabetes (11.6% of the population), with approximately 8.7 million undiagnosed (3.4%), and an estimated 97.6 million had pre-diabetes[3]. Diabetes affects people of all ages, socioeconomic statuses, and

ethnicities, highlighting how prevalent diabetes is across various populations and how vital this issue is today[3].

According to the Cleveland Clinic, diabetes occurs when the body has difficulty regulating its blood sugar (otherwise known as glucose) levels, resulting in high glucose levels[2]. This can happen when the pancreas does not produce enough insulin or does not produce any insulin at all, or when the body does not respond properly to insulin[2]. Over time, having these consistent, high glucose levels can have serious, long-term complications such as the risk of heart attacks, kidney disease, or nerve damage[2].

Although most forms of diabetes are chronic (lifelong), with the exception of reversible pre-diabetes and Type 2 diabetes, "which can be prevented, improved and sometimes even reversed," they all can be manageable with medications and/or lifestyle changes[1]. However, early diagnosis is vital to lowering one's risk of developing complications related to the disease. Brenda Sobleskey, RN and coordinator of diabetes education for McLaren Greater Lansing, emphasizes this point, stating, "[e]arly detection of diabetes can help prevent or delay complications[4]." Sobleskey emphasizes the importance of proactive measures, stating that managing blood sugar levels early and consistently is essential for reducing risks and improving the quality of life for those living with or at risk of diabetes[4].

Early diagnosis not only leads to lifestyle adjustments and more effective treatments for people with diabetes but also highlights the importance of predictive models for diabetes risk. DiabetesAI, which analyzes health-related data, such as "health-related risk behaviors, chronic health conditions, and access and use of health-care", serves as a valuable tool for both the general public and public health officials[5]. DiabetesAI can help individuals at risk of diabetes, allowing for actions that can delay the onset of diabetes and its future complications.

2 Related Works

Several studies have explored the use of machine learning techniques to predict and identify risk factors associated with diabetes. A 2014 study using the Behavioral Risk Factor Surveillance System (BRFSS) dataset built a predictive model for detecting Type 2 diabetes[6]. It analyzed data from 138,146 participants, including 20,467 diagnosed with Type 2 diabetes, and applied models like Support Vector Machines, Decision Trees, Logistic Regression, and Neural Networks[6]. The Neural Network model achieved the highest accuracy (82.4%) and AUC (0.7949), while the Decision Tree model had the highest sensitivity (51.6%), allowing for early diagnosis and intervention of diabetic patients[6]. Whereas, our project also focuses on diabetes risk prediction using similar models, but uses the 2015 BRFSS dataset and emphasizes utilizing feature like BMI and GenHlth for suggestions on how to improve diabetic patients' health.

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3 Dataset Overview

The Behavioral Risk Factor Surveillance System (BRFSS) is an annual health-related telephone survey conducted by the CDC[5]. It gathers responses from over "400,000 Americans about health-related risk behaviors, chronic health conditions, and the use of preventive services"[5]. For this project, we utilized a clean and consolidated CSV dataset from the 2015 BRFSS, available on Kaggle[5]. While the original dataset contained responses from 441,456 individuals and included 330 features, the cleaned and consolidated dataset included only 253,680 responses and 21 feature variables[5].

The dataset includes features like HighBP, BMI, MentHlth, Education, Fruits, and PhysActivity, with no missing values. Out of 22 features, 14 are binary, and the others are continuous or categorical. The target variable, Diabetes_012 had three classes: 0 for no diabetes or only during pregnancy, 1 for pre-diabetes, and 2 for diabetes. Additionally, there is class imbalance in the dataset (as shown in Figure 1), making it similar to real-world conditions and a good resource for studying the diabetes-related risk factors.

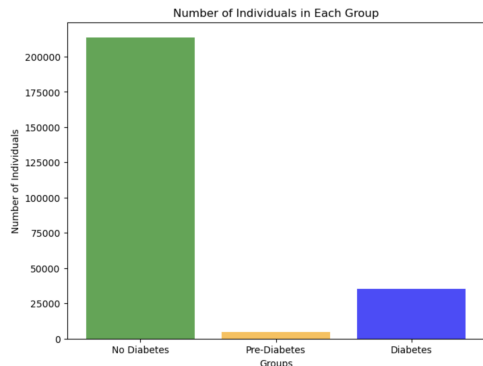


Figure 1: Distribution of the Class Imbalance for Diabetes

4 DiabetesAI Overview

DiabetesAI is a comprehensive model designed to address both identification and wellness aspects of diabetes management. It consists of two primary components:

- **Diabetes Prediction Model:** This part focuses on predicting whether a patient has diabetes or not based on various key features. The model then analyzes these factors to classify individuals into groups like diabetic (1) or non-diabetic (0), helping to identify at-risk patients earlier on.
- **Health Suggestion Component:** The second part of DiabetesAI aims to help individuals with diabetes by providing personalized health recommendations. This component evaluates individuals without diabetes by analyzing healthy ranges of key features like BMI and GenHlth. It then provides suggestions for individuals with diabetes, recommending how they can better manage their condition by staying in those healthy ranges.

5 Methods for the Diabetes Prediction Model

For the Diabetes Prediction Model, we used three machine learning algorithms to predict whether a patient has diabetes or not based on selected features such as BMI or GenHlth. The models we compared are Logistic Regression (originally intended as a baseline), Support Vector Machine, and Random Forest (to handle more complex, non-linear relationships).

5.1 Before Resampling

We implemented a Logistic Regression model to evaluate the performance without any resampling to address the class imbalance between non-diabetic (0), pre-diabetic (1), and diabetic features (2). The goal was to compare how the model performed in its original state before applying any techniques like oversampling, undersampling, or SMOTE.

Accuracy: 0.8482537054556922				
	precision	recall	f1-score	support
0.0	0.86	0.98	0.92	42795
1.0	0.00	0.00	0.00	944
2.0	0.54	0.18	0.27	6997
accuracy			0.85	50736
macro avg	0.47	0.39	0.40	50736
weighted avg	0.80	0.85	0.81	50736

Figure 2: Logistic Regression Model Performance Before Resampling

As shown in Figure 2, the model achieves a high accuracy overall, about 85%, which is expected due to the class imbalance—since the majority of data points belong to non-diabetes group, predicting "non-diabetic" would yield a correct prediction most of the time. However, the precision and recall for predicting diabetes (2) are relatively low, at around 54% and 18%, respectively. This means that 54% of the patients predicted to have diabetes actually do have diabetes, but the model only identifies 18% of the patients who actually have diabetes. For pre-diabetes (1), no correct predictions were made, and despite attempts to balance the classes, the results remained the same, leading us to combine pre-diabetes and diabetes, to better detect the risk of diabetes.

5.2 Data Pre-Processing

As mentioned earlier, since pre-diabetes could not be accurately predicted (with both its original dataset distribution and with resampling), we combined pre-diabetes and diabetes to better assess the risk of diabetes. However, before addressing the class imbalance, we first split the dataset into training and test sets (to make sure that the test set is a representative of the real-world data and to prevent data leakage). To balance the training data, we downsampled the non-diabetic cases to match the size of the diabetic group, as shown in Figure 3.

5.3 After Resampling

The next step in improving our model was selecting the most important features to identify diabetes for the training process. For Random Forest, we used tree-based embedded importance, which

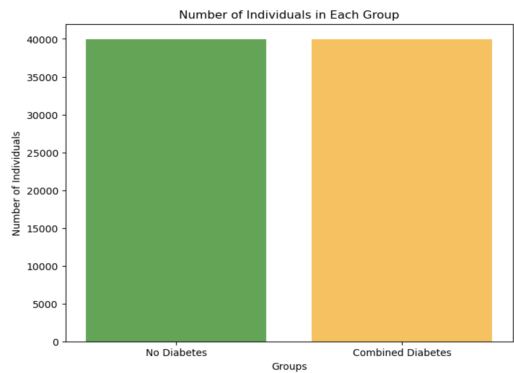


Figure 3: Resampled Distribution of Diabetic Versus Non-Diabetic Individuals

essentially evaluated how much each feature contributes to reducing impurity across all trees. For the Support Vector Machine model, we applied RFECV (Recursive Feature Elimination with Cross-Validation), which started with all features and iteratively removed the least important ones. For the Logistic Regression model, we determined feature importance using the model’s coefficients, which represented each feature’s contribution to predicting the diabetes. By averaging these coefficients across multiple iterations, we identified the features that were consistently the most influential. So, we were able to filter out the less relevant features such as Smoker, Fruits, Veggies, PhysActivity, AnyHealthcare, and Stroke to improve the model’s performance.

5.4 Results

Out of these three models, the Logistic Regression, contrary to our original expectations, performed the best. As shown in Figure 4, despite the overall average decreasing, the precision and recall scores significantly improved, achieving a precision score of 75% and recall score of 74%, in comparison to its pre-sampling performance of 54% and 18% respectively.

Accuracy: 0.7415421174410606				
	precision	recall	f1-score	support
0.0	0.75	0.72	0.74	7964
2.0	0.73	0.76	0.75	8027
accuracy			0.74	15991
macro avg	0.74	0.74	0.74	15991
weighted avg	0.74	0.74	0.74	15991

Figure 4: Logistic Regression Model Performance After Re-sampling and Feature Importance

6 Health Suggestion Component

For the Health Suggestion Component of the DiabetesAI model, we trained and tested a separate model using data from patients without diabetes to identify healthy ranges for key health indicators: BMI and GenHlth. This model was then applied to patients with

diabetes to provide suggestions for achieving healthier BMI and GenHlth values, helping them better manage their condition and improve overall health. Less relevant features, including CholCheck, HvyAlcoholConsump, AnyHealthcare, and Stroke, were excluded to refine the focus on these critical predictors.

6.1 BMI

Body Mass Index (BMI) is a measure of body weight relative to height, calculated by dividing a person’s weight (in kilograms) by the square of their height (in meters). To predict healthier BMI values for diabetic patients, we trained a Random Forest Regressor (because BMI is continuous) using data from 39,977 “healthy” or “non-diabetic” individuals as the training set. The model was able to achieve a Mean Squared Error (MSE) of 37.17 on the test set, which indicates that the predictions are not highly accurate and that there is room for improvement.

When the model was applied to diabetic patients, the model suggested BMI values centered around 25, which falls within the healthy BMI range as shown in Figure 5. However, the predicted values showed an average deviation of approximately 6 units, indicating that while the model captures general trends, further refinement is needed to improve the precision of the predictions for personalized health recommendations.

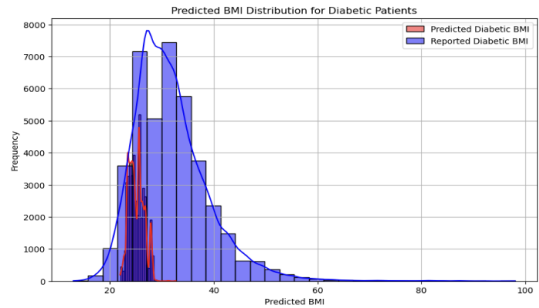


Figure 5: Predicted BMI Distribution for Diabetic Patients Versus Reported BMI Distribution

6.2 GenHlth

The GenHlth score is a scale from 1 to 5, where 1 represents “excellent” health and 5 represents “poor” health. For this prediction task, we used a Random Forest Classifier (because GenHlth is categorical from 1-5), training the model on data from healthy or “non-diabetic” individuals. The model was able to achieve a Mean Squared Error (MSE) of 0.89 on the test set, with a R² score of 0.16. While the MSE is relatively low, showing that the model’s predictions are not far off, the R² score suggests that the model explains only a small portion of the variance in the data.

When applied to diabetic patients, the model consistently predicted a GenHealth score of 3 (indicating “good” health). However, this score is too generalized and vague to provide meaningful suggestions for improving health. A more specific and precise quantification or qualification of health conditions is needed (what is considered “good” health versus “excellent” health, for example)

with a potential addition of other factors that may influence general health.

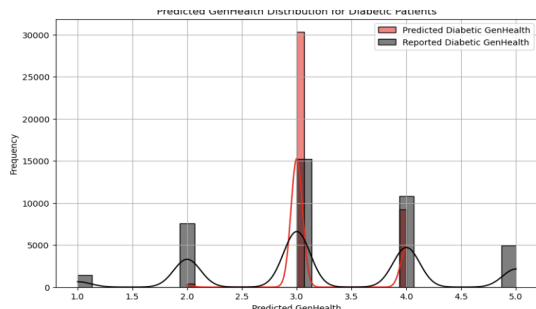


Figure 6: Predicted GenHlth Distribution for Diabetic Patients Versus Reported GenHlth Distribution

7 Discussion

The downsampling process in the data processing phase may introduce biases in the distribution of certain features used for training. Although we randomly selected samples from the non-diabetes group to match the size of the diabetes group, the inherent imbalances in the dataset could still lead to misrepresentation of feature distributions. For instance, while the education level does not show significant differences between non-diabetic and diabetic groups, it was deemed an important feature for diabetes prediction by the embedded Random Forest model. This could affect the reliability of the precision score for diabetes prediction, as it may reflect a biased distribution rather than a true representation of predictive power.

In the context of BMI prediction for diabetic patients, the high Mean Squared Error (MSE) in the model trained on healthy individuals suggests that some features in the training dataset may not accurately predict BMI values for diabetic individuals. This could indicate that the features captured in the training data are not sufficiently predictive of diabetes and may introduce confusion or noise into the model. Additionally, since BMI is strongly correlated with the presence of diabetes, it raises questions about whether behavioral features like income and education contribute meaningfully to diabetes prediction or simply add confusion to the model's final output. The vague recording of health-related features in the dataset also limits our ability to capture the underlying causes of diabetes, suggesting the need for more precise quantification and weighting of features to improve model performance.

8 Conclusion

In conclusion, survey questions from BRFSS can provide semi-accurate predictions of diabetes risk, with a precision of 0.74. However, to improve the model's accuracy, more data from diabetic patients is needed. Our analysis revealed that factors such as BMI, GenHealth, and MenHealth were strongly related to diabetes, highlighting the importance of these features in prediction models. However, for future data collection, it would be better to focus more on health conditions and physiological features specifically linked to diabetes. It's important to include more detailed medical histories (rather than a binary 0-1) of issues like comorbidities and

risk factors like hypertension and cholesterol levels, so it could improve model predictions.

Additionally, we found that certain factors, such as stroke, were excluded during feature selection, but they may require further investigation. Stroke indication could be a hint at individuals having cardiovascular problems, which may be relevant to diabetes risk.

To improve the model's performance, multiple ways are under our consideration. First, to address class balance, techniques like SMOTE (Synthetic Minority Over-sampling Technique) or class weighting could be utilized to help balance the distribution of diabetes and non-diabetes cases, allowing the model to better learn from the non-diabetic (minority) classes. Additionally, incorporating other ensemble learning techniques like Gradient Boosting or advanced approaches like neural networks may improve model robustness and prediction accuracy.

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